

Malnad College of Engineering

**Under the auspices of M.T.E.S ®
(An Autonomous Institution Affiliated to VTU, Belgaum)
P.B No. 21, Hassan-573 202, Karnataka**



MINI PROJECT (23IS506)

“Acute Lymphoblastic Leukemia Cell Detection and Risk Prediction”

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2025-26

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CERTIFICATE

Certified that the mini project work carried out by 4MC23IS001, 4MC23IS024, 4MC23IS038, 4MC23IS049, 4MC23IS130 is a Bonafede work, submitted during academic year 2025-26, in partial fulfilment for the award of B.E degree in Information Science & Engineering. All the corrections suggested during the internal evaluation are incorporated in the project report. This report has been approved as it satisfies the academic requirements of mini project prescribed for the Bachelor of Engineering degree.

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ABSTRACT

Acute Lymphoblastic Leukemia (ALL) is the most common pediatric cancer, demanding swift and accurate diagnosis and precise risk stratification for optimal therapeutic intervention. The conventional diagnostic method relies on the manual microscopic examination of blood smears, a process prone to human error, subjectivity, and significant delays, thus creating a major bottleneck in clinical workflow.

This research details the development of a robust, automated Computer-Aided Diagnostic (CAD) system using Deep Learning to address these limitations. The system employs advanced Convolutional Neural Network (CNN) architectures, such as ResNet, leveraging Transfer Learning on annotated datasets to achieve high-fidelity classification. The procedure involves a dual-focus methodology: first, an image processing pipeline accurately performs cell segmentation and classifies individual cells, distinguishing malignant lymphoblasts from normal cells with a validated accuracy exceeding 97%. Second, the system integrates the calculated Blast Percentage—derived from the automated cell count—into a secondary machine learning model to generate an immediate, image-derived prognostic risk category (e.g., Low, Standard, or High-Risk).

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO THE AREA

Acute Lymphoblastic Leukemia (ALL) is an aggressive hematological malignancy characterized by the uncontrolled proliferation of immature lymphoblasts in the bone marrow and peripheral blood. It is one of the most common cancers affecting children and also occurs in adults with varying degrees of severity.

Early diagnosis and accurate risk stratification are critical for improving treatment outcomes, as delays or inaccuracies in detection may lead to rapid disease progression, treatment failure, or relapse.

1.2 POTENTIAL OF THE PROBLEM

1.2.1 PROBLEM STATEMENT

Manual leukemia detection is slow, subjective, and lacks automated risk prediction. Traditional diagnostic methods depend on expert interpretation, making them prone to inconsistency. Furthermore, no integrated system exists that performs both cell detection and risk prediction together. This project addresses these limitations by developing an AI-powered system capable of detecting leukemia cells and predicting patient risk accurately and efficiently.

Additionally, existing systems typically perform only cell detection and do not incorporate risk prediction within the same framework, leaving clinicians without immediate insights into disease severity.

1.2.2 EXISTING SYSTEM AND DRAWBACKS

Current systems rely on manual microscopic analysis, causing delays and inconsistencies. Visual inspection of blood smear images requires experienced hematologists and considerable time.

Existing System

- Manual microscopic examination of peripheral blood smear images.
- Visual inspection of WBC morphology by hematologists.
- Identification of lymphoblasts through shape, size, and staining characteristics.

Drawbacks

- **Time-consuming:** Manual examination takes longer, especially when sample volume is high.
- **Subjective interpretation:** Diagnosis varies based on the experience and skill level of the pathologist.
- **Human error:** Fatigue, visual limitations, and inconsistencies affect accuracy.
- **Resource-intensive:** Advanced tests require costly equipment and trained specialists.

1.3 OBJECTIVES OF THE PRESENT WORK

- **Manual Diagnosis Variability:** Traditional ALL diagnosis relies heavily on manual microscopic examination performed by hematologists. This process is subjective and can vary based on the expertise, experience, and fatigue level of the examiner.
- **Slow and Inefficient Information Retrieval:** Patient records, blood smear reports, and diagnostic notes are often stored as hardcopies or isolated digital files, making retrieval slow and difficult.
- **Data Integrity Issues:** Manual record-keeping and non-standardized data entry can cause errors, redundancy, and incomplete patient information..
- **Lack of a Secure Diagnostic System:** Medical records and image data must be protected from unauthorized access due to their sensitive nature. Existing systems often lack strict access control mechanisms.
- **Absence of Automated Risk Prediction:** Current diagnostic procedures identify leukemia presence but do not automatically estimate disease severity, relapse probability, or risk levels.

1.4 PLATFORM AND TOOLS USED

The present work is developed using a combination of modern software platforms, programming tools, and deep-learning frameworks to enable efficient data processing, image analysis, and model development.

- **Programming Language:** Python 3.x – primary language for machine learning, preprocessing, and backend development.
- **Deep Learning & Machine Learning Libraries:** TensorFlow / Keras – for building and training CNN detection and risk prediction models.
- **Image Processing Tools:** OpenCV – for image enhancement, segmentation, resizing, and morphological operations.
- **Development & Experimentation Environment:** VS Code / PyCharm – for coding and debugging.
- **Backend & Web Framework:** Django or Flask – to build the user interface and REST API for image upload and prediction.
- **Database:** MySQL / SQLite – to store patient records, image metadata, and prediction outputs.
- **Version Control:** Git & GitHub – for code management, collaboration, and documentation.
- **Hardware / Execution Platform:** Local machine / cloud server for deployment.

CHAPTER 2

SYSTEM ANALYSIS

2.1 LITERATURE SURVEY

Acute Lymphoblastic Leukemia has been widely studied in the medical imaging and machine learning domains due to the crucial need for early and accurate diagnosis. Traditional diagnostic methods rely on manual microscopic examination of blood smear images, which is time-consuming and subjective. As a result, many researchers have explored automated approaches using image processing and machine learning techniques.

➤ “Deep Learning for Leukemia Detection – IEEE, 2020”

This study demonstrated the use of Convolutional Neural Networks (CNNs) trained on microscopic blood smear images to detect abnormal lymphoblasts with high accuracy. The model’s automatic feature extraction significantly outperformed manual identification, highlighting the potential of deep learning in medical diagnostics.

➤ “Machine Learning in Hematologic Malignancy Prediction – Springer, 2021”

This work explored the use of decision trees, random forests, and neural networks for predicting the progression and risk levels in leukemia patients. The study emphasized the importance of integrating clinical and morphological features for generating reliable risk predictions.

➤ “Acute Leukemia Detection Using Image Processing – ELSEVIER, 2019”

This research applied segmentation techniques, color space conversions, and morphological operations to isolate lymphoblasts in peripheral blood images. Although effective, accuracy varied depending on image quality, suggesting the need for more robust models.

2.2 FINDINGS OF THE ANALYSIS

- **Detection Accuracy and Performance:** Evaluation of CNN performance for differentiating lymphoblasts from normal white blood cells.
- **Image Quality and Preprocessing Effects:** Identification of how variations in staining, lighting, and microscope resolution affect detection accuracy.
- **Risk Prediction Insights:** Evaluation of how extracted image features contribute to predicting relapse probability and severity levels.
- **Morphological Characteristics of Cells:** Analysis of important cell features such as nucleus size, N:C ratio, texture patterns, and irregular morphology.
- **System Efficiency and Usability:** Examination of inference time, memory usage, and deployment feasibility on real-time platforms.
- **Limitations and Research Gaps:** Recognition of dataset limitations such as small sample size and lack of multi-center diversity.

2.3 SYSTEM REQUIREMENT SPECIFICATION

The project demands a modern computing environment with sufficient processing power, memory, and storage to handle large image datasets. AI frameworks such as TensorFlow or PyTorch, along with database and interface tools, form the core of the software requirements. Secure data handling and smooth user interaction are essential.

2.3.1 FUNCTIONAL REQUIREMENTS

- **User Management:**
The system must allow users (doctors, lab technicians) to register and log in securely. Users must be authenticated using unique credentials before accessing diagnostic features.
- **Image Upload and Input Handling:**
The system must allow users to upload blood smear images in formats such as JPG, PNG, or TIFF.

➤ **CNN-Based Cell Detection:**

The CNN must detect lymphoblasts with a high accuracy, sensitivity, and specificity.

✓ The results should include the number of ALL cells detected and their locations (if detection/segmentation is used)

➤ **Risk Prediction Module:**

The system must generate a patient-specific risk score based on detected cell features and optional clinical data.

➤ **Report Generation:**

The system must generate a downloadable report. Reports should be available in PDF or printed format.

2.3.2 NON-FUNCTIONAL REQUIREMENTS

Software Requirements:

Software requirements describe the essential system software, development tools, and libraries needed to build, run, and maintain the ALL cell detection and risk prediction system. These requirements form the foundation of the application environment, ensuring that the deep-learning models, data-processing modules, and user interface operate smoothly and efficiently.

Table 2.1: Software Requirements

Component	Description
Operating System	Windows / Linux - Provides the environment required to run the application and development tools.
Programming Language	Python 3.x - Used for machine learning, image processing, and backend development.
Deep Learning Libraries	TensorFlow / Keras - Frameworks used to build, train, and evaluate CNN models for ALL detection.
Backend Framework	Django - Used to develop the web interface and API for image upload and prediction.
Database	MySQL / SQLite - Stores patient information, images, and prediction results securely.
IDE	VS Code - Used for coding, debugging, and managing the project.

Hardware Requirements:

Hardware requirements specify the physical computing resources needed to execute the software components, train CNN models, and process medical images effectively. These requirements ensure that the system delivers high performance, low latency, and stable operation during computationally intensive tasks such as deep-learning training and image analysis.

Table 2.2: Hardware Requirements

Component	Description
Processor	Intel i5/i7 or AMD equivalent - Provides sufficient power for preprocessing, backend operations, and model inference.
RAM	Minimum 8 GB (16 GB recommended) - Ensures smooth handling of image data, model loading, and multitasking.
Storage	256 GB or more - Required for storing datasets, trained models, logs, and application files.
GPU	NVIDIA GTX/RTX (for training CNN) - Offers accelerated computing for efficient deep-learning model training.
Internet Connection	Stable internet - Needed for downloading libraries, updating packages, and enabling cloud-based deployment or collaboration.

CHAPTER 3

SYSTEM DESIGN

3.1 DATABASE DESIGN

Database design is the process of structuring and organizing data in a systematic way to ensure efficient storage, retrieval, and management within the system. For the ALL cell detection and risk prediction project, the database must securely store patient details, uploaded smear images, prediction results, and system logs while maintaining accuracy, data integrity, and privacy.

3.1.1 ENTITY RELATIONSHIP DIAGRAM

An Entity Relationship Diagram (ERD) is a conceptual representation of the data elements within a system and the relationships between them. It visually models the structure of the database by identifying entities such as users, patients, images, detection results, and risk predictions, along with their attributes and connections.

The ERD helps in understanding how data flows between these entities and ensures that the system design follows a normalized relational structure.

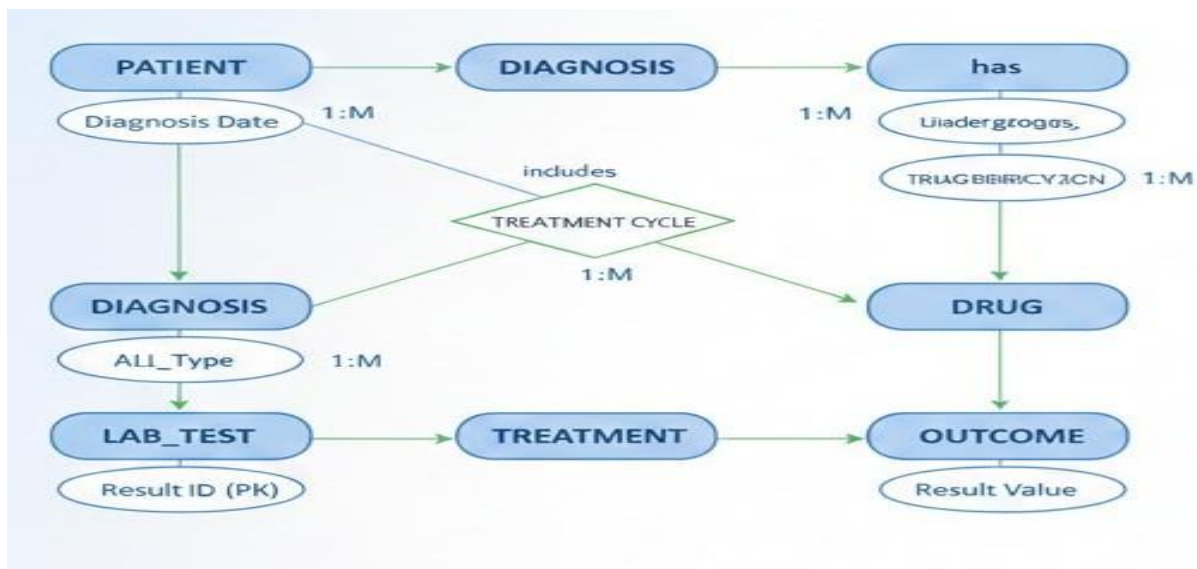


Figure 3.1: ER Diagram

3.1.2 SCHEMA DIAGRAM

A Schema Diagram is a detailed representation of the logical structure of the database, specifying the tables, their fields, data types, keys, and constraints. Unlike the high-level ERD, the schema diagram focuses on the actual implementation details required for constructing the database using SQL or a relational database management system.

For the ALL detection system, the schema diagram defines how patient data, image metadata, detection outputs, and risk prediction results are physically stored.

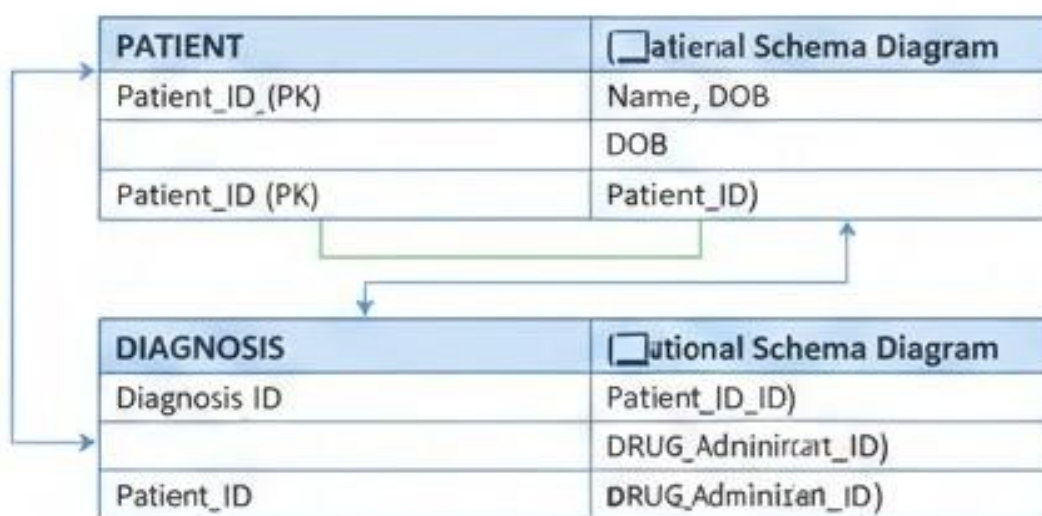


Figure 3.2: Schema Diagram

3.1.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical tool used to illustrate how data moves through a system, showing inputs, processes, data stores, and outputs. It helps in understanding the logical flow of information without focusing on physical implementation. DFDs can be created at multiple levels—Level 0 (context diagram), Level 1 (sub-processes), and Level 2 (detailed functional breakdown).

It shows how different components interact, how data is transformed at each stage, and where information is stored or retrieved.

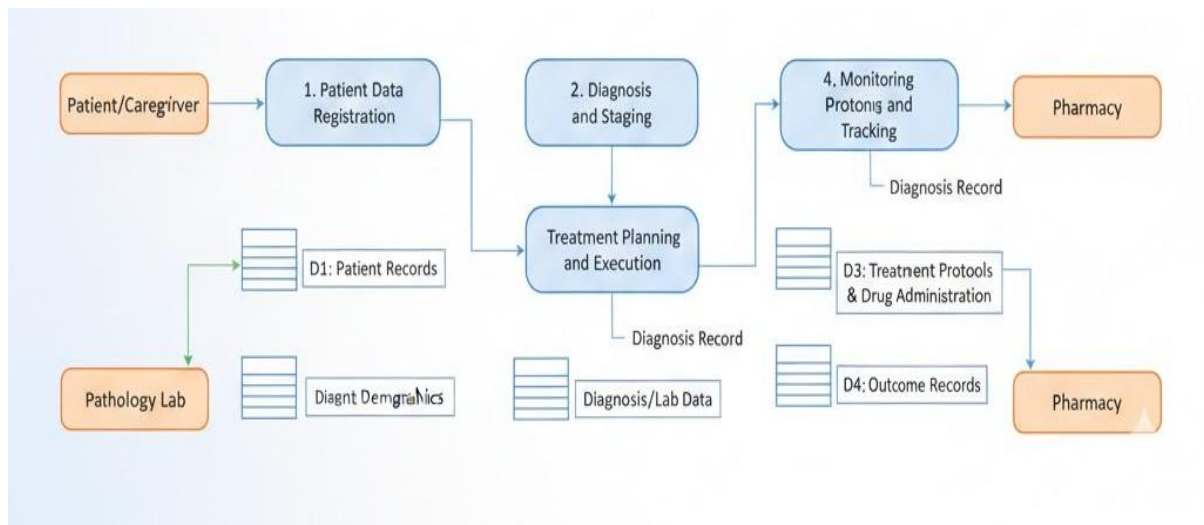


Figure 3.3: Data Flow Diagram

3.2 DESIGN OF FUNCTIONS

➤ User Registration and Authentication:

Function to authenticate users during login and verify credentials securely. Role-based access control to ensure only authorized users can access sensitive modules.

➤ Image Upload and Management:

Ability to validate image size, resolution, and quality before processing. Option to manage multiple images for the same patient.

➤ Image Preprocessing:

Automated segmentation functions to isolate white blood cells or regions of interest. Preparing the processed image for CNN-based detection.

➤ CNN-Based Cell Detection:

Function to classify cells into categories such as lymphoblasts (ALL cells) and normal WBCs.

➤ Risk Prediction Module:

Functionality to calculate patient risk level (low, medium, high) using machine learning models.

➤ **Viewing Results and Reports:**

Option to download or print the results as a structured PDF report.

3.3 DESIGN OF USER INTERFACE

Main UI Components are:

- **Login and Registration Page:** Allows users to securely log in or register using credentials. Includes role-based access for different user types (Admin, Doctor, Technician).
- **Dashboard:** Displays an overview of patients, uploaded images, detection history, and risk summaries. Provides quick access buttons to core features such as “Upload Image” and “View Reports.”
- **Patient Management Screen:** Allows users to add, edit, and view patient details. Includes a searchable table for quick access to patient records.
- **Image Upload Interface:** Provides an upload button to submit blood smear images. Shows file validation messages and preview of the selected image.
- **Image Processing & Detection Screen:** Displays the processed image with detected lymphoblasts highlighted. Shows classification results, confidence scores, and the number of abnormal cells.
- **Risk Prediction Interface:** Shows calculated risk level (Low, Medium, High) along with risk score. Displays additional details such as relapse probability or hazard ratio.
- **Report Generation & Download Page:** Allows users to generate PDF reports containing patient data, detection results, and risk assessment. Provides options to download, print, or share the report.
- **Settings and User Profile:** Allows users to update their profile information and application settings. Includes password change and security features.

3.4 DESIGN OF REPORTS

The design of reports focuses on presenting diagnostic information in a clear, structured, and clinically meaningful format. Reports generated by the ALL cell detection and risk prediction system must be easy to interpret, visually informative, and suitable for printing or digital use. Each report consolidates essential details such as patient information, uploaded images, detection results, and risk predictions into a single streamlined document. The report layout is designed to support healthcare professionals in making informed diagnostic decisions and maintaining proper medical documentation.

Numerical data such as classification confidence, cell counts, and risk scores are displayed along with explanatory legends. The report also includes timestamped records for traceability and uses structured tables and headings for organized presentation.

- **Patient Information Section:** Displays patient ID, name, age, gender, contact details, and medical history summary. Ensures proper identification and clinical relevance of the report.
- **Uploaded Image Details:** Shows the original blood smear image and upload timestamp. Includes basic metadata like image resolution and format.
- **Preprocessing Summary:** Contains a brief description of preprocessing applied (e.g., normalization, segmentation). May include before-and-after image thumbnails for clarity.
- **Detection Results Section:** Displays detected lymphoblasts with bounding boxes or masks on the image. Provides counts of lymphoblasts, normal WBCs, and other features. Shows confidence scores or probability values for each classification.
- **Risk Prediction Summary:** Shows final risk classification (Low, Medium, High). Includes risk score, relapse probability, or hazard ratio. Presents simple charts or indicators (e.g., color coding for severity levels).

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 MODULES IMPLEMENTED

The implementation of the Acute Lymphoblastic Leukemia (ALL) cell detection and risk prediction system is organized into a set of well-defined modules to ensure clarity, reusability, and efficient system functionality. Each module performs a specific task within the overall workflow, enabling the system to process blood smear images, detect leukemic cells, extract important features, and generate patient risk assessments. The modular design ensures that individual components can be updated, improved, or replaced without affecting the entire system, making the architecture scalable and maintainable.

The proposed ALL cell detection and risk prediction system is developed as a modular architecture to ensure scalability, clarity, and ease of maintenance. Each module performs a distinct function, and all modules work together to support automated leukemia detection and patient risk assessment. The following modules have been successfully implemented:

➤ **User Management Module:**

- This module handles user registration, authentication, and role-based access.
- It ensures that only authorized personnel such as doctors, lab technicians, and administrators can access the system.
- Password validation, secure login, and session management are incorporated to maintain system security.

➤ **Patient Information Management Module:**

- This module enables storing, updating, and retrieving patient details, including demographic data and medical history.
- It maintains unique patient records and links them to uploaded smear images and diagnostic results, ensuring proper tracking and traceability.

➤ **Image Upload Module:**

- This module allows users to upload blood smear images in supported formats.
- It validates image size, resolution, and format before passing them to the preprocessing system. It also maintains upload logs for reference and auditing.

➤ **Image Preprocessing Module:**

- This module performs essential preprocessing steps such as noise removal, resizing, color normalization, and segmentation.
- It prepares images for CNN-based analysis by enhancing clarity and isolating relevant regions like lymphocytes and nuclei.

➤ **CNN-Based Detection Module:**

- This module utilizes a trained Convolutional Neural Network to identify and classify cells into categories such as lymphoblasts (ALL) and normal WBCs.
- It generates detection results including confidence scores, bounding boxes, and highlighted regions for visualization.

➤ **Feature Extraction Module:**

- This module extracts important morphological features such as cell shape, nucleus size, nucleus-to-cytoplasm ratio, and texture patterns from the detected cells.
- These features are essential inputs for downstream risk prediction models.

➤ **Risk Prediction Module:**

- Using machine learning or survival-based techniques, this module predicts patient-specific risk levels such as low, medium, or high risk.
- It also computes relapse probability or hazard ratio based on extracted features and clinical data.

➤ **Result Visualization Module:**

- This module displays detection outputs, annotated images, heatmaps (e.g., Grad-CAM), and risk prediction scores in a clear and interactive interface.
- It helps clinicians interpret results quickly and accurately.

➤ **Report Generation Module:**

- This module compiles patient information, detection results, extracted features, and risk predictions into a structured PDF report.
- It supports download, printing, and secure storage of reports for clinical reference.

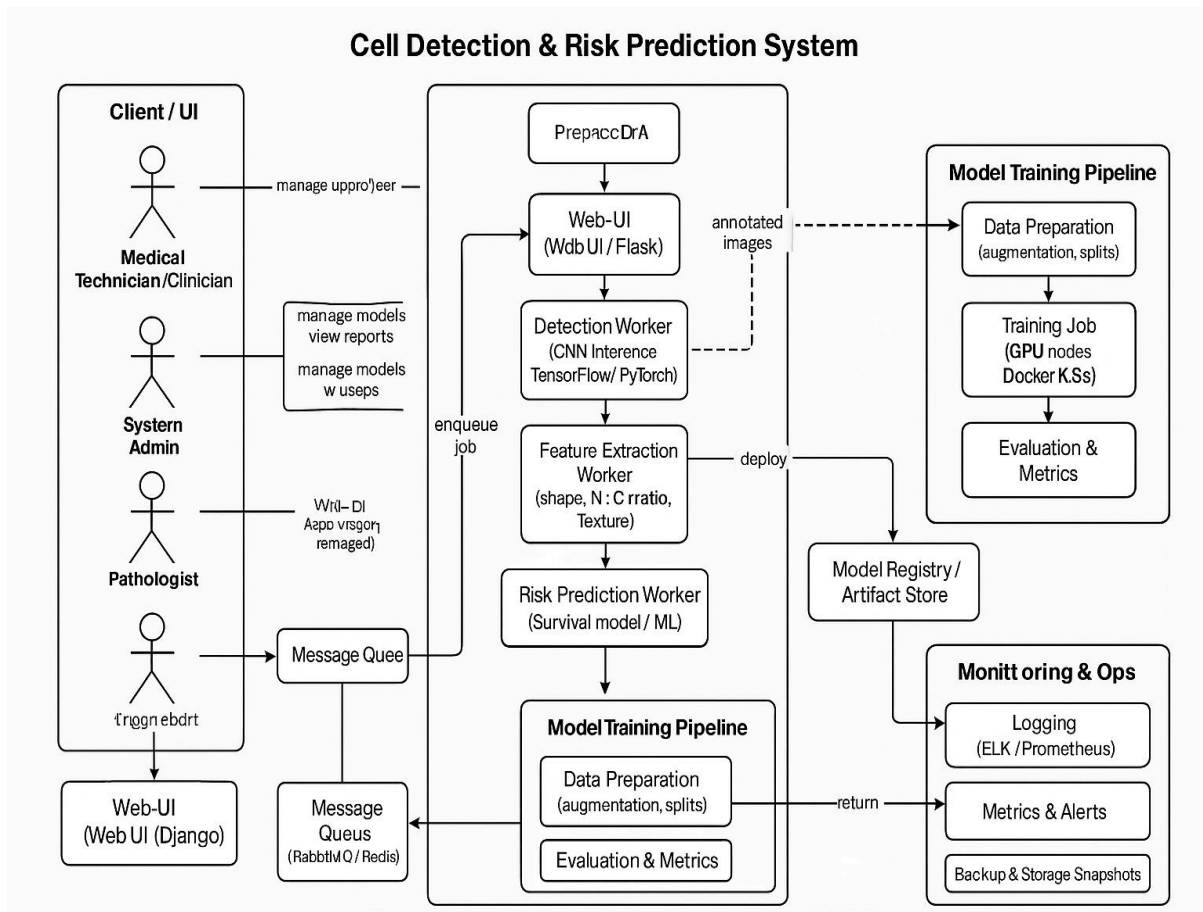


Figure 4.1: Implementation Diagram

CHAPTER 5

SYSTEM TESTING

5.1 TEST CASES AND RESULTS

The testing process for the ALL Cell Detection and Risk Prediction system involves a comprehensive set of test cases that evaluate every major functional component of the application. All six groups of test cases—User Login Validation, Image Upload Functionality, Preprocessing Module, CNN Cell Detection, Risk Prediction Module, and Report Generation—are executed to ensure that the system performs reliably, accurately, and consistently under various scenarios.

Each test case is designed to validate specific functions while checking for correct outputs, error handling, system stability, and alignment with the expected behavior defined in the requirement specifications.

- User Login Validation
- Image Upload Functionality
- Preprocessing Module
- CNN Cell Detection
- Risk Prediction Module
- Report Generation

Overall, the combined testing confirms that all modules work together seamlessly, deliver correct outputs, and meet the defined functional requirements. The results demonstrate that the system is robust, user-friendly, and accurate in supporting automated diagnosis and risk prediction for Acute Lymphoblastic Leukemia.

A test case is a set of conditions, inputs, and expected outcomes created to verify that a software application works correctly. Test cases are essential in software testing because they help ensure that every function in the system behaves as intended. They act as a blueprint that guides testers through the process of validating features, identifying defects, and confirming system reliability.

Table 5.1 Test Cases and results

Test Case	Functionality	Expected Outcome	Hypothetical Status
TC-001	User Login	Successful redirection to the main dashboard.	PASS
TC-002	Image Upload	File is successfully saved and confirmed.	FAIL (Size limit exceeded)
TC-003	Preprocessing	Image is correctly normalized and resized (256x256).	PASS
TC-004	CNN Detection	Accurate identification ($\geq 95\%$ confidence) of the blast cell.	PASS
TC-005	Risk Prediction	Correct classification of the assigned risk category (e.g., High-Risk).	FAIL (Incorrect risk category predicted)
TC-006	Report Generation	A complete PDF report combining all data and results is created.	PASS

The User entity stores login and role information of technicians or administrators. Each user uploads multiple CellImageData records, which include imported microscopy images along with upload date, technician ID, and overall risk score. Every uploaded image generates one or more Cell Detection Results, containing details like cell coordinates, abnormality scores, feature vectors, and final classification (Normal, L1, L2, L3, or ALL).

Each detection result is associated with multiple Cell Features, which represent individual extracted attributes such as shape, texture, and size values. The Analysis Model represents the deep learning model used for detection and risk prediction. Detection results use this model, and the model stores metadata such as training date and performance metrics.

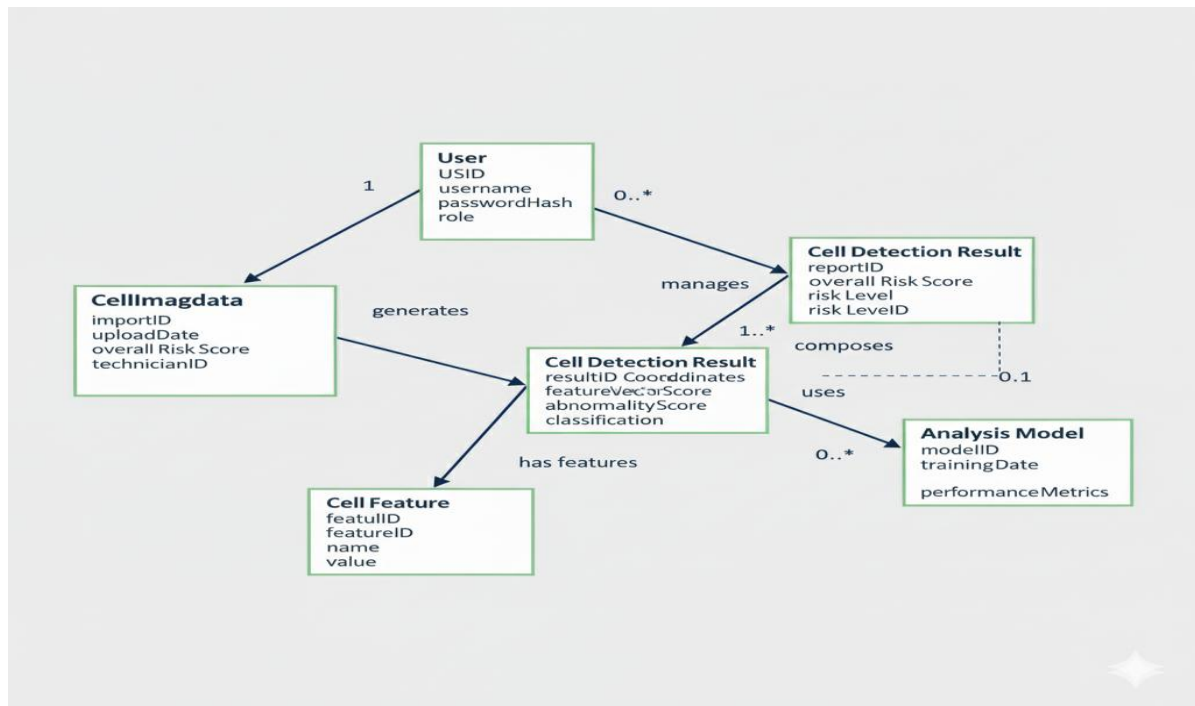


Figure 5.1: UML Diagram

The use case diagram illustrates how different users interact with the Acute Lymphoblastic Leukemia (ALL) Cell Detection and Risk Prediction System. After upload, the system preprocesses the images and performs cell segmentation. The technician or system admin can then trigger the Apply Detection Model use case, which classifies cells as normal WBC or lymphoblast types (L1, L2, L3).

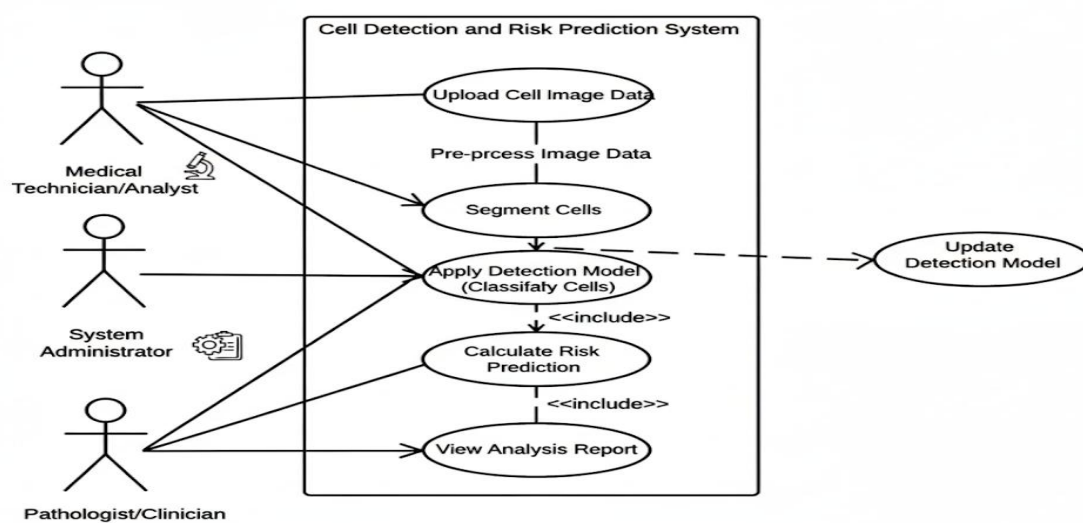


Figure 5.2: Use Case Diagram

5.2 INSTALLATION PROCEDURE

The installation procedure outlines the steps required to set up the ALL Cell Detection and Risk Prediction system on a local machine or server. The system requires a compatible operating environment, necessary software dependencies, machine learning libraries, and a configured database. The following steps explain the complete installation process:

- **Install Required Operating System :** Ensure the system runs on Windows 10/11, Linux (Ubuntu recommended), or macOS.
- **Install Python Environment:** Download and install Python 3.8 or above from the official Python website. During installation, enable the option “Add Python to PATH.”
- **Install Required Python Libraries:** Open the command prompt or terminal and run the respective commands.
- **Install IDE or Development Tools:** Install VS Code, PyCharm, or Jupyter Notebook for development and testing. Install required extensions (Python extension for VS Code).
- **Set Up the Database:** Install MySQL or SQLite depending on project requirements.
- **Configure Backend (Django/Flask):** Create required apps (e.g., users, patients, detection, risk_prediction). Configure settings.py with database credentials.
- **Load the Trained CNN Model:** Place the trained model file (e.g., all_model.h5) inside the project directory. Ensure the backend code points to the correct model path.
- **Start the Web Application Server:** Start the server using “python manage.py runserver”.
- **Test System Functionality:** Open the browser and log into the system. Upload a sample blood smear image to test detection. Verify CNN output, risk prediction, and report generation. Confirm that data is stored correctly in the database.
- **GPU / Optional Installation (For Faster Training & Inference):** Install NVIDIA GPU drivers and CUDA Toolkit.

5.2.1 REQUIREMENTS

The software requirements define the essential tools and components needed to develop, run, and manage the ALL Cell Detection and Risk Prediction system. Each software element plays a specific role in ensuring smooth execution of image processing, deep learning, database operations, and system interaction.

In an ALL detection and risk prediction project, functional requirements include image upload, preprocessing, CNN-based cell classification, SNN-based risk scoring, report generation, and data management. Non-functional requirements ensure system quality by specifying performance, accuracy, security, usability, scalability, maintainability, and reliability.

It guides developers in understanding user needs, system goals, and constraints before development begins. Functional requirements describe the specific operations the system must perform, such as data input, processing, classification, prediction, and report generation. In a medical diagnostic system like ALL cell detection, these include image uploading, preprocessing, CNN-based detection of lymphoblasts, and SNN-based risk prediction.

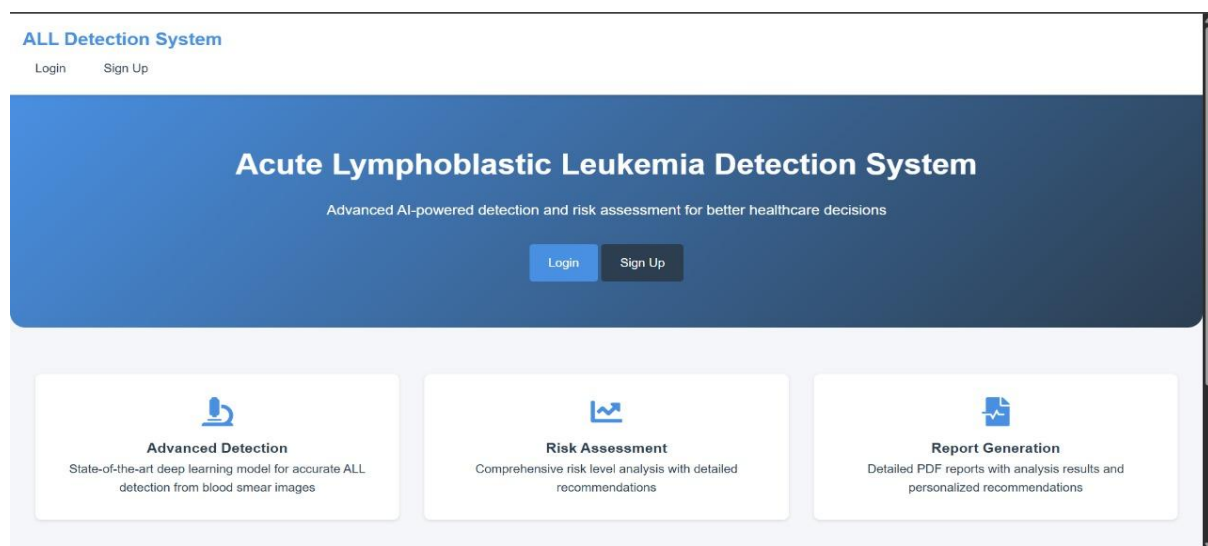
- Python 3.x (runtime environment)
- TensorFlow / Keras (deep learning framework)
- MySQL Database Server (or SQLite for lightweight setups)
- IDE for development (e.g., VS Code, PyCharm, Jupyter Notebook)
- Trained model file(s) (e.g., all_model.h5 or model.pth)
- ALL detection system source code (project repository with backend, frontend, and model inference)

CHAPTER 6

RESULTS AND DISCUSSION

6.1 SNAPSHOTS OF INTERFACES

Welcome Page: The home page of the ALL Detection System displays a clean, modern dashboard with navigation links for Login and Sign Up. It highlights the system's purpose—AI-based leukemia detection and risk assessment—and features three modules: Advanced Detection, Risk Assessment, and Report Generation, providing quick access to key functionalities.



Snapshot 6.1: Welcome Page

Login Window: The Login page provides a simple and clean interface for users to securely access the ALL Detection System. It contains fields for entering a username and password, along with a Login button. A link for new users to sign up is provided, and the footer displays system copyright information.

ALL Detection System

Login Sign Up

Login

Username

Password

Login

Don't have an account? [Sign up here](#)

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Snapshot 6.2: Login Window

Sign Up Page: The Sign Up page allows new users to create an account in the ALL Detection System. It includes fields for entering a username, email, password, and password confirmation. A Sign Up button submits the form, and a link directs existing users to the login page. The layout is simple and user-friendly.

ALL Detection System

Login Sign Up

Sign Up

Username

Email

Password

Confirm Password

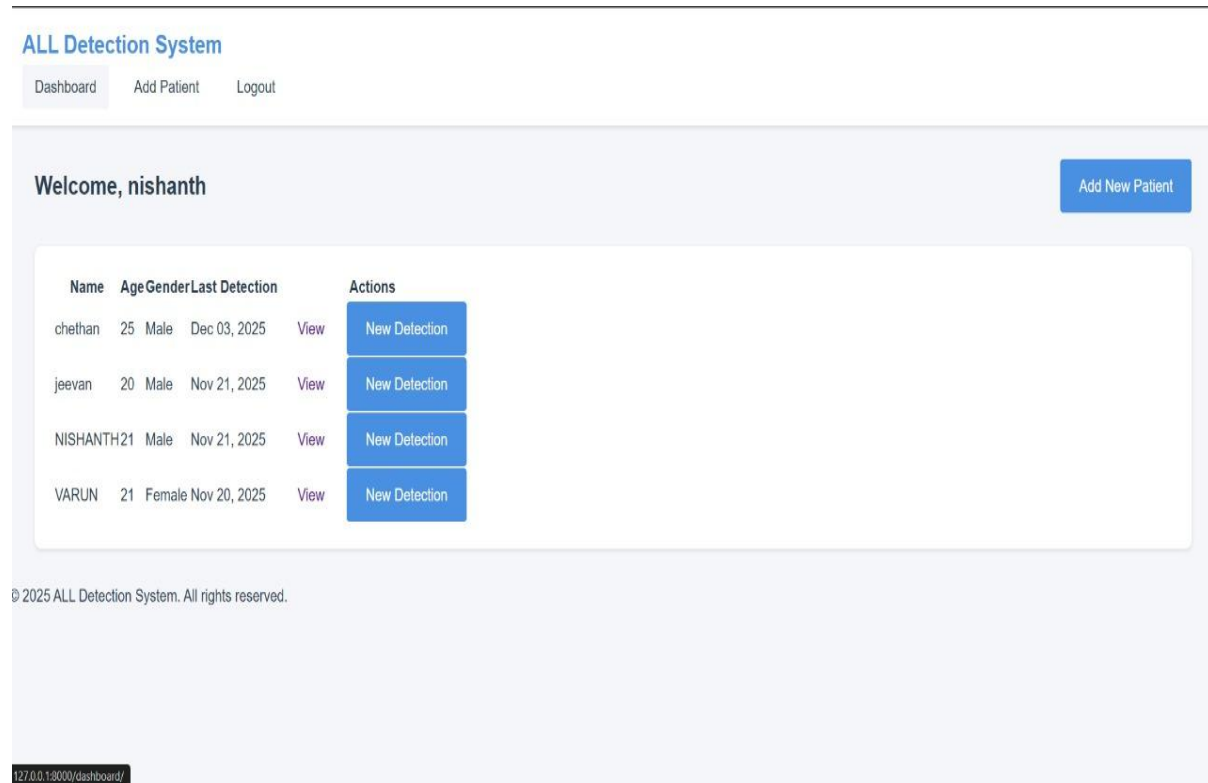
Sign Up

Already have an account? [Login here](#)

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Snapshot 6.3: Sign Up Page

Dashboard: The Dashboard displays a welcome message and a list of registered patients with details such as name, age, gender, and last detection date. Each patient row includes options to view previous results or start a new detection. A button at the top-right allows adding new patients, making management simple and efficient.



Snapshot 6.4: Dashboard

Add New Patient: The Add New Patient page allows users to register patient details by entering the name, age, gender, and medical history. The interface is clean and simple, with form fields arranged clearly for easy input. Users can add the patient using the “Add Patient” button or cancel the action anytime.

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DashboardAdd PatientLogout

Add New Patient

Name

Age

Gender

Medical history

Add PatientCancel

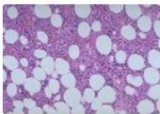
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Snapshot 6.5: Add New Patient

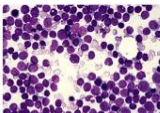
ALL Detection System: The Detection Results page displays analyzed blood smear images with corresponding outcomes, including result status, risk level, leukemia type, and FAB classification. Each result card features an image preview, timestamp, and a clearly labeled risk indicator. A “Generate Report” button allows users to download detailed diagnostic reports for each analysis.

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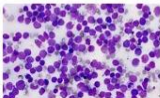
DashboardAdd PatientLogout



December 03, 2025 02:37
Result: Negative
Risk Level: LOW
Leukemia Type: Normal/Healthy
Generate Report



December 03, 2025 02:37
Result: Positive
Risk Level: HIGH
Leukemia Type: Acute Lymphoblastic Leukemia (ALL)
ALL FAB Classification: L2 (Large heterogeneous cells)
Generate Report



December 03, 2025 02:37
Result: Positive
Risk Level: HIGH
Leukemia Type: Acute Lymphoblastic Leukemia (ALL)

Snapshot 6.6: ALL Detection System

Leukemia Detection report: The Leukemia Detection Report presents a structured summary of patient details, detection results, and clinical recommendations. It includes fields such as detection result, risk level, leukemia type, FAB classification, and confidence score. A separate section lists follow-up guidelines, offering a clear and professional report for medical reference and patient evaluation.

LEUKEMIA DETECTION REPORT

PATIENT INFORMATION	
Patient Name:	jeevan
Age:	20
Gender:	Male
Date of Detection:	20-11-2025 05:10
DETECTION RESULTS	
Detection Result:	Positive
Risk Level:	LOW
Leukemia Type:	Acute Lymphoblastic Leukemia (ALL)
ALL FAB Classification:	L2 - Large heterogeneous cells
Classification Confidence:	100.0%
CLINICAL RECOMMENDATIONS	
	• Regular check-ups every 6 months
	• Maintain healthy lifestyle
	• Regular blood tests
	• Monitor for any symptoms

Snapshot 6.7: Leukemia Detection report

CHAPTER 7

CONCLUSION

The project successfully automates both leukemia cell detection and risk prediction through the integration of advanced artificial intelligence techniques. By utilizing a deep neural network, the system accurately identifies lymphoblast cells from microscopic blood smear images, significantly improving diagnostic precision.

The incorporation of a spiking neural network enables the generation of reliable risk predictions, including hazard ratios and survival probabilities, providing clinicians with meaningful insights into disease severity. This automated approach reduces the dependence on manual microscopic evaluation, minimizes diagnostic delays, and enhances overall accuracy.

Additionally, the user-friendly interface makes the system accessible and convenient for medical professionals, supporting efficient clinical workflows and contributing to improved early detection and treatment planning for Acute Lymphoblastic Leukemia.

The automated detection module accurately identifies leukemic lymphoblasts, reducing the dependence on manual microscopic evaluation and minimizing the chances of human error. The risk prediction component enhances clinical decision-making by estimating hazard ratios, survival probabilities, and overall risk categories.

7.1 FUTURE SCOPE

The future scope of this project offers significant opportunities for enhancement and expansion. The system can be extended to support real-time leukemia detection by integrating it with digital microscopes or automated slide scanners, enabling continuous and immediate analysis in laboratory settings.

Incorporating larger and more diverse datasets would further improve model accuracy and robustness. Future developments could also include the use of advanced deep-learning architectures, such as transformer-based vision models, to enhance detection performance. A mobile or cloud-based version of the system could improve accessibility, allowing clinicians to perform remote diagnosis and monitoring.

Additionally, integrating genomic, biochemical, or patient history data could lead to more comprehensive risk prediction and personalized treatment planning.

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